MA1522 Linear Algebra for Computing
L2 Lesson 4



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Semester 1, 2026/27

Elementary Matrices

An elementary matrix $\boldsymbol{E} = (e_{ij})_{n\times n}$ is one that can be obtained from $\boldsymbol{I}_n$ by an elementary row operation.

$R_a \leftrightarrow R_b$ $(a \neq b)$:
$$e_{ij} = \begin{cases} 1, & \text{if } i = j \notin \{a, b\}; \\ 1, & \text{if } \{i, j\} = \{a, b\}; \\ 0, & \text{otherwise.} \end{cases}$$

$R_a + cR_b$ $(a \neq b, c \neq 0)$:
$$e_{ij} = \begin{cases} 1, & \text{if } i = j; \\ c, & \text{if } (i, j) = (a, b); \\ 0, & \text{otherwise.} \end{cases}$$

cR_a $(c \neq 0)$:
$$e_{ij} = \begin{cases} 1, & \text{if } i = j \neq a; \\ c, & \text{if } i = j = a; \\ 0, & \text{otherwise.} \end{cases}$$

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Elementary Matrices vs Elementary Row Operations

Theorem

Performing an elementary row operation to a matrix $\boldsymbol{A}$ is equivalent to pre-multiplying $\boldsymbol{A}$ by the corresponding elementary matrix.

Reversing Elementary Row Operations

Reversing an elementary row operation is an elementary row operation of the same type.

$$R_i \leftrightarrow R_j \xrightarrow{\text{reversing}} R_i \leftrightarrow R_j$$

$$R_i + cR_j \xrightarrow{\text{reversing}} R_i - cR_j$$

$$cR_i \xrightarrow{\text{reversing}} \frac{1}{c}R_i$$

Question 1

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Let $\boldsymbol{A} = \begin{pmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{pmatrix}$.

a. Find the reduced row-echelon form $\boldsymbol{R}$ of $\boldsymbol{A}$.

b. Write $\boldsymbol{A}$ in the form $\boldsymbol{E}_1\boldsymbol{E}_2\cdots\boldsymbol{E}_k\boldsymbol{R}$, where $\boldsymbol{E}_1, \ldots, \boldsymbol{E}_k$ are elementary matrices.

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Solution.

a.

$$\boldsymbol{A} \xrightarrow{R_2+2R_1} \begin{pmatrix} -1 & 3 & -4 \\ 0 & 10 & -7 \\ -4 & 2 & -9 \end{pmatrix} \xrightarrow{R_3-4R_1} \begin{pmatrix} -1 & 3 & -4 \\ 0 & 10 & -7 \\ 0 & -10 & 7 \end{pmatrix} \xrightarrow{R_3+R_2} \begin{pmatrix} -1 & 3 & -4 \\ 0 & 10 & -7 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{-R_1} \begin{pmatrix} 1 & -3 & 4 \\ 0 & 10 & -7 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{\frac{1}{10}R_2} \begin{pmatrix} 1 & -3 & 4 \\ 0 & 1 & -\frac{7}{10} \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1+3R_2} \begin{pmatrix} 1 & 0 & \frac{19}{10} \\ 0 & 1 & -\frac{7}{10} \\ 0 & 0 & 0 \end{pmatrix} =: \boldsymbol{R}$$

Question 1(b): Solution

Write $\boldsymbol{A}$ in the form $\boldsymbol{E}_1\boldsymbol{E}_2\cdots\boldsymbol{E}_k\boldsymbol{R}$, where $\boldsymbol{E}_1,\ldots,\boldsymbol{E}_k$ are elementary matrices.

From part (a),

$$\boldsymbol{A}\xrightarrow{R_2+2R_1}\xrightarrow{R_3-4R_1}\xrightarrow{R_3+R_2}\xrightarrow{-R_1}\xrightarrow{\frac{1}{10}R_2}\xrightarrow{R_1+3R_2}\boldsymbol{R},$$

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Thus $\boldsymbol{A}=\boldsymbol{E}_1\boldsymbol{E}_2\boldsymbol{E}_3\boldsymbol{E}_4\boldsymbol{E}_5\boldsymbol{E}_6\boldsymbol{R}$, where

$$\boldsymbol{E}_1=\begin{pmatrix}1&0&0\\-2&1&0\\0&0&1\end{pmatrix},\ \boldsymbol{E}_2=\begin{pmatrix}1&0&0\\0&1&0\\4&0&1\end{pmatrix},\ \boldsymbol{E}_3=\begin{pmatrix}1&0&0\\0&1&0\\0&-1&1\end{pmatrix},$$

$$\boldsymbol{E}_4=\begin{pmatrix}-1&0&0\\0&1&0\\0&0&1\end{pmatrix},\ \boldsymbol{E}_5=\begin{pmatrix}1&0&0\\0&10&0\\0&0&1\end{pmatrix},\ \boldsymbol{E}_6=\begin{pmatrix}1&-3&0\\0&1&0\\0&0&1\end{pmatrix}.$$

Question 1(b): Solution

Write $\boldsymbol{A}$ in the form $\boldsymbol{E}_1\boldsymbol{E}_2\cdots\boldsymbol{E}_k\boldsymbol{R}$, where $\boldsymbol{E}_1,\ldots,\boldsymbol{E}_k$ are elementary matrices.

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Thus $\boldsymbol{A} = \boldsymbol{E}_1\boldsymbol{E}_2\boldsymbol{E}_3\boldsymbol{E}_4\boldsymbol{E}_5\boldsymbol{E}_6\boldsymbol{R}$, where

$$\boldsymbol{E}_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \boldsymbol{E}_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{pmatrix}, \quad \boldsymbol{E}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix},$$

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Inverse of a Matrix

A square matrix $\boldsymbol{A}$ is said to be **invertible** if and only if there exists a matrix $\boldsymbol{B}$ of the same size such that $\boldsymbol{AB} = \boldsymbol{I} = \boldsymbol{BA}$, in which case $\boldsymbol{B}$ is said to be the **inverse of $\boldsymbol{A}$**, denoted as $\boldsymbol{A}^{-1}$.

Theorem

Let A and B be square matrices of order n.

1. *If A is invertible, then*
   a. *A^{-1} is invertible, and $(A^{-1})^{-1} = A$.*
   b. *A^T is invertible, and $(A^T)^{-1} = (A^{-1})^T$.*
   c. *if $c \in \mathbb{R}$, $c \neq 0$, then cA is invertible, with $(cA)^{-1} = \frac{1}{c}A^{-1}$.*
2. *If A and B are both invertible, then AB is invertible, with $(AB)^{-1} = B^{-1}A^{-1}$.*

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Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be square matrices of order n.

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   a. *$\boldsymbol{A}^{-1}$ is invertible, and $(\boldsymbol{A}^{-1})^{-1} = \boldsymbol{A}$.*
   b. *$\boldsymbol{A}^T$ is invertible, and $(\boldsymbol{A}^T)^{-1} = (\boldsymbol{A}^{-1})^T$.*
   c. *if $c \in \mathbb{R}$, $c \neq 0$, then $c\boldsymbol{A}$ is invertible, with $(c\boldsymbol{A})^{-1} = \frac{1}{c}\boldsymbol{A}^{-1}$.*
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Equivalent Statements of Invertibility

Theorem

Let $\boldsymbol{A}$ be a square matrix. The following statements are equivalent:

1. *$\boldsymbol{A}$ is invertible.*
2. *$\boldsymbol{A}^T$ is invertible.*
3. *There is a matrix $\boldsymbol{B}$ such that $\boldsymbol{BA} = \boldsymbol{I}$.*
4. *There is a matrix $\boldsymbol{B}$ such that $\boldsymbol{AB} = \boldsymbol{I}$.*
5. $\mathrm{RREF}(\boldsymbol{A}) = \boldsymbol{I}$.
6. *$\boldsymbol{A}$ can be expressed as a product of elementary matrices.*
7. *The homogeneous system $\boldsymbol{Ax} = \boldsymbol{0}$ has only the trivial solution.*
8. *For any $\boldsymbol{b}$, the system $\boldsymbol{Ax} = \boldsymbol{b}$ has a unique solution.*

Question 2

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True or false: If $\boldsymbol{A}$ and $\boldsymbol{B}$ are square matrices of the same size, and $\boldsymbol{AB}$ is invertible, then both $\boldsymbol{A}$ and $\boldsymbol{B}$ are invertible.

Question 2

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True or false: If $\boldsymbol{A}$ and $\boldsymbol{B}$ are square matrices of the same size, and $\boldsymbol{AB}$ is invertible, then both $\boldsymbol{A}$ and $\boldsymbol{B}$ are invertible.

Ans: True

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Ans: True

Proof.

Let $\boldsymbol{C} = (\boldsymbol{AB})^{-1}$. Then

$$\boldsymbol{I} = \boldsymbol{C}(\boldsymbol{AB}) = (\boldsymbol{CA})\boldsymbol{B},$$

so that $\boldsymbol{B}$ is invertible.

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Ans: True

Proof.

Let $\boldsymbol{C} = (\boldsymbol{AB})^{-1}$. Then

$$\boldsymbol{I} = \boldsymbol{C}(\boldsymbol{AB}) = (\boldsymbol{CA})\boldsymbol{B},$$

so that $\boldsymbol{B}$ is invertible.

Similarly,

$$\boldsymbol{I} = (\boldsymbol{AB})\boldsymbol{C} = \boldsymbol{A}(\boldsymbol{BC}),$$

so that $\boldsymbol{A}$ is invertible. □

Question 3

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For each of the following matrices, determine if it is invertible, and if so, find its inverse.

a. $\begin{pmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{pmatrix}$

b. $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 0 & 1 \\ 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 1 \end{pmatrix}$

(Cf. Pg 82–85 and Pg 115 of Chapter 2 Lecture Slides.)

Question 3: Solution

(a)

$$\left(\begin{array}{ccc|ccc} -1 & 3 & -4 & 1 & 0 & 0 \\ 2 & 4 & 1 & 0 & 1 & 0 \\ -4 & 2 & -9 & 0 & 0 & 1 \end{array}\right) \xrightarrow[\text{Elimination}]{\text{Gaussian}} \left(\begin{array}{ccc|ccc} -1 & 3 & -4 & 1 & 0 & 0 \\ 0 & 10 & -7 & 2 & 1 & 0 \\ 0 & 0 & 0 & -2 & 2 & 1 \end{array}\right)$$

Since the third column is non-pivot, the matrix is not invertible.

(b)

$$\left(\begin{array}{cccc|cccc} 1 & 2 & 3 & 4 & 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 2 & 0 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[\text{Jordan}]{\text{Gauss-}} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & 0 & \frac{1}{2} & \frac{1}{4} & -1 & -\frac{1}{4} \\ 0 & 0 & 1 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{4} & 0 & -\frac{3}{4} \end{array}\right)$$

Thus, the matrix is invertible with inverse $\begin{pmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} & -1 & -\frac{1}{4} \\ -\frac{1}{2} & 0 & \frac{1}{2} & 1 \\ \frac{1}{2} & -\frac{1}{4} & 0 & -\frac{3}{4} \end{pmatrix}$.

Question 3: Solution

a

$$\left(\begin{array}{ccc|ccc} -1 & 3 & -4 & 1 & 0 & 0 \\ 2 & 4 & 1 & 0 & 1 & 0 \\ -4 & 2 & -9 & 0 & 0 & 1 \end{array}\right) \xrightarrow[\text{Elimination}]{\text{Gaussian}} \left(\begin{array}{ccc|ccc} -1 & 3 & -4 & 1 & 0 & 0 \\ 0 & 10 & -7 & 2 & 1 & 0 \\ 0 & 0 & 0 & -2 & 2 & 1 \end{array}\right)$$

Since the third column is non-pivot, the matrix is not invertible.

b

$$\left(\begin{array}{cccc|cccc} 1 & 2 & 3 & 4 & 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 2 & 0 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[\text{Jordan}]{\text{Gauss-}} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & 0 & \frac{1}{2} & \frac{1}{4} & -1 & -\frac{1}{4} \\ 0 & 0 & 1 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{4} & 0 & -\frac{3}{4} \end{array}\right)$$

Thus, the matrix is invertible with inverse $\begin{pmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} & -1 & -\frac{1}{4} \\ -\frac{1}{2} & 0 & \frac{1}{2} & 1 \\ \frac{1}{2} & -\frac{1}{4} & 0 & -\frac{3}{4} \end{pmatrix}$.

Next Lesson ...

We will explore **Sections 2.7–2.9**.
Please watch the video clips and review lecture slides for these sections before then.